

Divide Multi-Digit Numbers

Lesson 1-4

Name:

Date:

Class:

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

12 dots

$$12 \div 3 = 4$$

In $156 \div 12 = 13$, the dividend is 156 — it is the total being split up

Dividend

Write the definition:



$$12 \div 3 = 4$$

In $156 \div 12 = 13$, the divisor is 12 — it is the size of each group

Divisor

Write the definition:



$$12 \div 3 = 4$$

In $156 \div 12 = 13$, the quotient is 13 — there are 13 groups

Quotient

Write the definition:



$$7 \div 3 = 2 \text{ R}1$$

$17 \div 5 = 3 \text{ R} 2$ means 3 groups of 5 with 2 left over

Remainder

Write the definition:



$$12 \div 3 = 4$$

1,344 $\div$ 12: first $12 \times 100 = 1,200$, then $12 \times 12 = 144$. Quotient = $100 + 12 = 112$

Partial quotients

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can divide multi-digit whole numbers and explain the meaning of the quotient and remainder.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Dividend

Divisor

Quotient

Remainder

Partial quotients

1

In $84 \div 7$, the number 84 being divided is the _____.

2

In $84 \div 7$, the number 7 that we divide by is the _____.

3

The answer to a division problem is called the _____.

4

When a number does not divide evenly, the amount left over is the _____.

5

Breaking division into easier chunks and adding the pieces uses _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is $936 \div 12$?

- A. 78
- B. 87
- C. 68
- D. 76

Step 1 $12 \times 78 = 936$.

Step 2 You can verify: $12 \times 70 = 840$, $12 \times 8 = 96$, $840 + 96 = 936$.

 **Answer:** A. 78

 We do — together

Solve this one with your class using the same steps.

Problem: What is $2,485 \div 5$?

- A. 497
- B. 487
- C. 507
- D. 496

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: What is $756 \div 9$?

- A. 84
- B. 83
- C. 94
- D. 86

Show your work:

| Try It

Solve on your own. Check the answer key when you are done.

1. Path 1: The first arcade door shows $4,928 \div 7$. Which quotient lights the correct path?

- A. 704
- B. 74
- C. 740
- D. 703 r7

Show your work:

2. Path 2: A side corridor displays $6,045 \div 9$. Which door is correct?

- A. 671 R 6
- B. 671
- C. 672 R 3
- D. 67 R 6

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A school has 1,680 pencils to distribute equally to 24 classrooms. Show how to solve this using partial quotients. Explain each step and check your answer by multiplying.

Sentence starter: $1,680 \div 24 = \underline{\quad}$. First I used $24 \times \underline{\quad} = \underline{\quad}$, then $24 \times \underline{\quad} = \underline{\quad}$. The partial quotients add to $\underline{\quad}$. I checked by multiplying: $24 \times \underline{\quad} = 1,680$.

Show your work:

Reflect — Exit Ticket

What is $2,352 \div 16$?

- A. 147
- B. 137
- C. 157
- D. 146

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Dividend
2. **Notes 2:** Divisor
3. **Notes 3:** Quotient
4. **Notes 4:** Remainder
5. **Notes 5:** Partial quotients
6. **We Try (notes):** A. $497 - 5 \times 497 = 2,485$. *Check: $5 \times 400 = 2,000$, $5 \times 90 = 450$, $5 \times 7 = 35$.
 $2,000 + 450 + 35 = 2,485$.*
7. **You Try (notes):** A. $84 - 9 \times 84 = 756$. *Check: $9 \times 80 = 720$, $9 \times 4 = 36$, $720 + 36 = 756$.*
8. **Try It 1:** A. $704 - 7 \times 704 = 4,928$ exactly, so the quotient is 704 with no remainder.
9. **Try It 2:** A. $671 \text{ R } 6 - 9 \times 671 = 6,039$, and $6,045 - 6,039 = 6$, so $6,045 \div 9 = 671 \text{ R } 6$.
10. **Exit Ticket:** A. $147 - 16 \times 147 = 2,352$. *Check: $16 \times 100 = 1,600$, $16 \times 40 = 640$, $16 \times 7 = 112$.
 $1,600 + 640 + 112 = 2,352$.*